Example 11.30

Evaluate the following integrals :

(i) $\int 2x\sqrt{1+x^2}\,dx$ (ii) $\int e^{-x^2}x\,dx$ (iii) $\int \frac{\sin x}{1+\cos x}dx$

(iv) $\int \frac{1}{1+x^2}dx$ (v) $\int x(a-x)^8\,dx$

Solution

(i) $\int 2x\sqrt{1+x^2}\,dx$

Putting $1+x^2=u$, then $2x\,dx=du$

$$\int 2x\sqrt{1+x^2}\,dx = \int \sqrt{u}\,du$$

$$= \int u^{\frac{1}{2}}du = \frac{u^{\frac{3}{2}}}{\frac{3}{2}}+c = \frac{2}{3}u^{\frac{3}{2}}+c = \frac{2}{3}(1+x^2)^{\frac{3}{2}}+c\,.$$

(ii) $\int e^{-x^2}x\,dx$

Putting $x^2=u$ then $2x\,dx=du$

Therefore, $$\int e^{-x^2}x\,dx = \int e^{-u}\frac{du}{2}$$

$$= \frac{1}{2}\int e^{-u}du = \frac{1}{2}(-e^{-u})+c = -\frac{1}{2}e^{-u}+c = -\frac{1}{2}e^{-x^2}+c.$$

(iii) $\int \frac{\sin x}{1+\cos x}dx$

Putting $1+\cos x=u$, then $-\sin x\,dx=du$

Therefore, $$\int \frac{\sin x}{1+\cos x}dx = \int \frac{-du}{u} = -\log|u|+c = -\log|1+\cos x|+c.$$

(iv) $\int \frac{1}{1+x^2}dx$

Putting $x=\tan u$, then $dx=\sec^2 u\,du$

$$\int \frac{1}{1+x^2}dx = \int \frac{\sec^2 u}{1+\tan^2 u}du = \int \frac{\sec^2 u}{\sec^2 u}du = \int du = u+c$$

$$\int \frac{1}{1+x^2}dx = \tan^{-1}x+c.$$

(v) $\int x(a-x)^8\,dx$

Putting $u=a-x$, then $du=-dx$

$$\int x(a-x)^8\,dx = \int x(a-x)^8\,dx$$

$$= \int (a-u)(u)^8(-du)$$

$$= \int \left(-a(u)^8+u^9\right)du$$

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